

Rudra Gupta

Machine Learning & Computer Vision Engineer

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SUMMARY

Computer Science undergraduate focused on Machine Learning and Computer Vision, with hands-on experience building real-time vision pipelines in Python using OpenCV and MediaPipe and data analysis workflows with Pandas, NumPy and scikit-learn. Comfortable taking a model from experimentation to a deployed, user-facing application.

EDUCATION

Vellore Institute of Technology, Bhopal *B.Tech, Computer Science and Engineering*

Sep 2023 – May 2027

CGPA: **9.15 / 10.0**

TECHNICAL SKILLS

Languages: Python, Java, SQL
ML & CV: Machine Learning, Computer Vision, Feature Engineering, Model Evaluation, Real-Time Inference
Libraries: NumPy, Pandas, scikit-learn, OpenCV, MediaPipe, Matplotlib, Seaborn
Data: MySQL, PostgreSQL, MongoDB
Deployment: Flask/Django APIs, AWS, Git, GitHub

PROJECTS

TrackMySpend: Expense Analytics & Spend Forecasting Application [\[GitHub\]](#) 2025

Python, Pandas, NumPy, Matplotlib, Seaborn, MySQL, Twilio, Tkinter

- Built a desktop analytics application for student finances that maintains separate online and offline balances against a savings goal, storing a categorised, multi-year transaction ledger in MySQL with date, category and type filters.
- Trained a least-squares linear regression over month-aggregated spending to forecast the next period's outflow, surfacing the prediction beside a running average on the profile dashboard and plotting the fitted trend line with its equation for interpretability.
- Developed an analysis workspace of three Matplotlib/Seaborn views embedded in Tkinter – entry-wise spend trend, month-wise credited-versus-debited comparison.
- Integrated a Twilio WhatsApp channel so expenses can be logged by message, with a sync routine that parses incoming entries and reconciles them into the database behind authenticated accounts and encrypted, environment-managed credentials.

AI Gesture Control System: Real-Time Hand-Tracking Interface [\[GitHub\]](#) 2025

Python, OpenCV, MediaPipe, NumPy

- Built a real-time computer vision pipeline that streams webcam frames through MediaPipe's 21-point hand landmark model and classifies six distinct gestures to drive system actions – volume, media play/pause, mouse click, photo capture and screenshot.
- Engineered geometric features from landmark coordinates (fingertip-to-joint positions and inter-landmark distances) for the gesture classifier, and added temporal smoothing to suppress false triggers between frames.
- Ran simultaneous face and hand tracking per frame with an on-screen overlay rendering live landmarks and the currently recognised gesture as immediate visual feedback.
- Structured the system into modular detection, gesture-handling, configuration and UI components, keeping inference responsive on standard CPU-only hardware.

Decision Review System (DRS): Cricket Video-Analysis Simulator [\[GitHub\]](#) 2024

Python, OpenCV, imutils, Pillow, Tkinter

- Developed a third-umpire simulator performing frame-level video analysis with OpenCV – multi-angle replay, frame-by-frame stepping and variable-speed slow/fast motion review of contentious deliveries.
- Implemented LBW impact prediction by detecting and tracking the ball across frames from a user-selected stump region, then extrapolating its trajectory to estimate whether it would strike the wickets.
- Handled a library of 12+ delivery clips with camera-angle switching, decoding, resizing and colour-space conversion of frames through OpenCV, imutils and Pillow at interactive playback rates.
- Delivered the analysis through a Tkinter interface with broadcast-style Pending / Out / Not Out decision states and manual override controls.

CERTIFICATIONS

- **Oracle Certified Foundations Associate - Agentic AI** *Oracle University, 2026*
- **ServiceNow Virtual Internship Program** *ServiceNow University / SmartBridge, 2026*
- **Introduction to Machine Learning** *IIT Madras, 2025*
- **AWS Technical Essentials** *AWS Training & Certification, 2025*